

UNIT – 1

OBJECTIVE :

To familiarize the students with the fundamentals of thermodynamics . so it will be useful to study about the thermodynamics in detail manner.

REAL GAS EQUATION:

At high pressures, the gases start to deviate from ideal –gas behavior. So the deviation should be accounted. For accounting this deviation a factor called compressibility is introduced. So the state equation for real gases is given as,

$$pv = ZRT$$

where, Z = compressibility factor.

The compressibility factor, $Z = pv / RT$

The compressibility factor can also be expresses as

$$Z = V_{\text{actual}} / V_{\text{Ideal}}$$

Where, V_{actual} - Specific volume of real gases.

V_{Ideal} – Specific Volume of Ideal gases.

For ideal gases $Z = 1$. But for Real gases, Z is greater than or less than unity.

IDEAL GAS:

An ideal gas is an imaginary substance that obeys the relation $pV = RT$ or $pV = RT$. so, the equation of state is also known as ideal- gas equation. At low pressures and high temperature, the density of a gas decreases. At that time, the gas behaves as an ideal gas. A factor is introduced to account the deviation of ideal gas behavior known as **compressibility factor**. It means a measure of deviation of the ideal gas behaviour

At normal condition H_2 , O_2 , N_2 and air behave as ideal gas.

“A Perfect gas may be defined as a state of a substance, whose evaporation from its liquid state is complete and strictly obeys all the gas laws under all conditions of temperature and pressure.

- No Real or actual gas strictly obey the gas laws.
- Real gases which are ordinarily difficult to liquify, such as oxygen, N_2 , H_2 and air.

Difference Between Ideal Gas & Real Gas

IDEAL GAS	REAL GAS
The behaviour of ideal gases is SIMPLE.	The behaviour of ideal gases is COMPLEX.
Mass is considered as a “point mass”. It means that the particle is extremely small where its mass is almost zero. So, it does not have volume.	Real gases are made up of molecules or atoms which occupies some space. It has a definite volume even they are extremely small.
Only air is considered as ideal gas.	All gases are considered as real gas except air.
The pressure of ideal gas is more	The pressure of a real gas is lesser as compared to ideal gas
It follows $pV = nRT$. This is the equation of ideal gas	It does not follow idea gas equation. So the equation of state for real gases become $p_v = ZRT$ or $(p+a/V^2) (V-b) = nRT$.

LAW OF PERFECT GASES:

Types of Laws :

1. Boyle's Law : (*Robert Boyle in 1662*)

“The Absolute pressure of a given mass of a perfect gas varies inversely as its volume, when the temperature remains constant”

Mathematically,

$$P \propto 1/v \quad \text{or} \quad Pv = \text{constant}$$

$$\text{Then, } P_1V_1 = P_2V_2 = P_3V_3 = \dots\dots\dots = \text{const.}$$

2. Charles' Law : (*Frechman Jacques A.C. Charles in 1787*)

“The volume of a given mass of a perfects gas varies directly as its absolute temperature, when the absolute pressure remains constant.”

Mathematically

$$V \propto T$$

$$V/T = \text{Constant}$$

$$\text{Then, } V_1/T_1 = V_2/T_2 = V_3/T_3 = \dots\dots\dots = \text{const.}$$

3. Gay-Lussac Law :

“The Absolute pressure of a given mass of a perfect gas varies directly as its absolute temperature, when the volume remains constant”

Mathematically,

$$P \propto T$$

$$P/T = \text{constant}$$

$$\text{then, } P_1/T_1 = P_2/T_2 = P_3/T_3 = \dots\dots\dots = \text{Const.}$$

4. Joule's Law :

“The change of internal energy of a perfect gas is directly proportional to the change of temperature.”

Mathematically

$$dE \propto dT$$

$$dE = mc \, dt$$

constant proportionality known as specific heat

$$dE = mc (T_2 - T_1)$$

Temp. of a given mass m of a gas changes from T_1 to T_2 , then Internal Energy changes from E_1 to E_2 and change in internal energy ($E_2 - E_1$) will be same irrespective of the manner how the pressure (P) and volume (v) of the gas have changed.

5. Regnaut's Law :

“The two specific heats C_p and C_v of a gas do not change with the change of temperature and pressure.”

Eg., c_p and c_v always constant.

It is only an approximate one, because the specific heats of a gas vary with temperature

6. Avogadro's Law :

- “avogadro's law states that equal volumes of different perfect gases at the same temperature and pressure contain equal number of molecules.
- It can also be stated, “the volume of one gram mole of all gases at the pressure of 760mm Hg and temperature of 0°C is same and equal to 22.4 litres.

$$\text{Kg mole of a gas} = 22.4 \times 10^{-3} \text{ cm}^3 = 22.4 \text{ m}^3.$$

7. Characteristic Gas Equation:

The general gas equation for ideal gas is given by

$$pV / T = \text{Constant}$$

Where, p – Pressure in **N/ m³**

V – Volume in **m³**

T – Temperature **°C**

Taking R as Constant

$$pV / T = R$$

$$pV = RT$$

If the mass of gas ' m ' is considered, the equation of state becomes

$$pV = mRT$$

This equation is known as Characteristic gas equation.

EQUATION OF STATE FOR REAL GAS:

1. Vander Waal's Equation :

The equation of state for real gases is given by

$$\left[P + \frac{a}{v^2} \right] (v - b) = RT$$

For ideal gas, the constants a and b are zero. The values of a and b are dependent on the type of fluid or gases used.

If molar volume is considered in the analysis, the equation of state becomes

$$\left[P + \frac{a}{\bar{V}^2} \right] (\bar{V} - b) = \bar{R}T$$

Where, $\bar{V}$ = molar volume

$\bar{R}$ = Universal Gas Constant = 8.314 Nm/ KgmolK

The value of a & b is theoretically determined by

$$a = \frac{27 R^2 (T_c)^2}{64 P_c} \quad \& \quad b = \frac{R T_c}{8 P_c}$$

Where, T_c – critical Temperature

P_c - Critical Pressure.

2. Redlich - kwong equtaion

$$p = \frac{RT}{V_m - b} - \frac{a}{\sqrt{T} V_m (V_m + b)}$$

$$a = \frac{0.42748 R^2 T_c^{5/2}}{P_c} \quad ; \quad b = \frac{0.08664 R T_c}{P_c}$$

This equation gives good result at high pressure and is fairly accurate for temperature above the critical value

3. Dieterici equation :

$$P(V - b) = RT e^{-a/RTV}$$

It does not give accurate value of the properties

PROPERTIES OF PURE SUBSTANCES

DEFINITION OF THE PURE SUBSTANCE

A pure substance is a system which is (i) homogeneous in composition, (ii) homogeneous in chemical aggregation, and (iii) invariable in chemical aggregation.

- “*Homogeneous in composition*” means that the composition of each part of the system is the *same* as the composition of *every other part*. “*Composition*” means the relative proportions of the chemical elements into which the sample can be analysed. It does not matter how these elements are combined.

For example in Fig.1 system (a), comprising steam and water, is homogeneous in composition, since chemical analysis would reveal that hydrogen and oxygen atoms are present in the ratio 2:1 whether the sample be taken from the steam or from the water. The same is true of system (b)

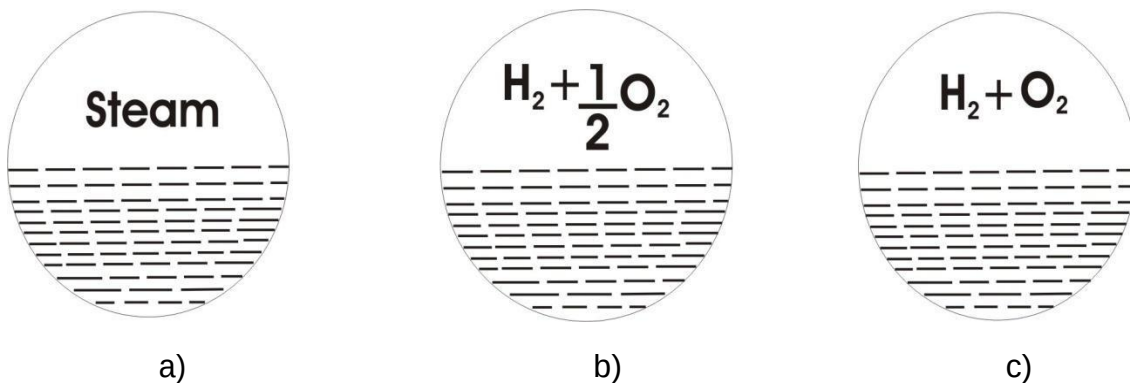


Fig.1 Illustrating the definition of a pure substance.

Containing uncombined hydrogen and oxygen gas in the atomic ratio 2: 1 in the upper part, and water in the lower part. System (c) however, is not homogeneous in composition, for the hydrogen and oxygen are present in the ratio 1:1 in the upper part, but in the ratio 2:1 (as water) in the lower part.

— “*Homogeneous in chemical aggregation*” means that the chemical elements must be combined chemically in the same way in all parts of the system. Consideration of Fig.1 again shows that the system (a) satisfies this condition also; for steam and water consist of identical molecules. System (b) on the other hand is not homogeneous in chemical aggregation since in the upper part of the system the hydrogen and oxygen are not combined chemically (individual atoms of H and O are not uniquely associated), whereas in the lower part of the system the hydrogen and oxygen are combined to form water.

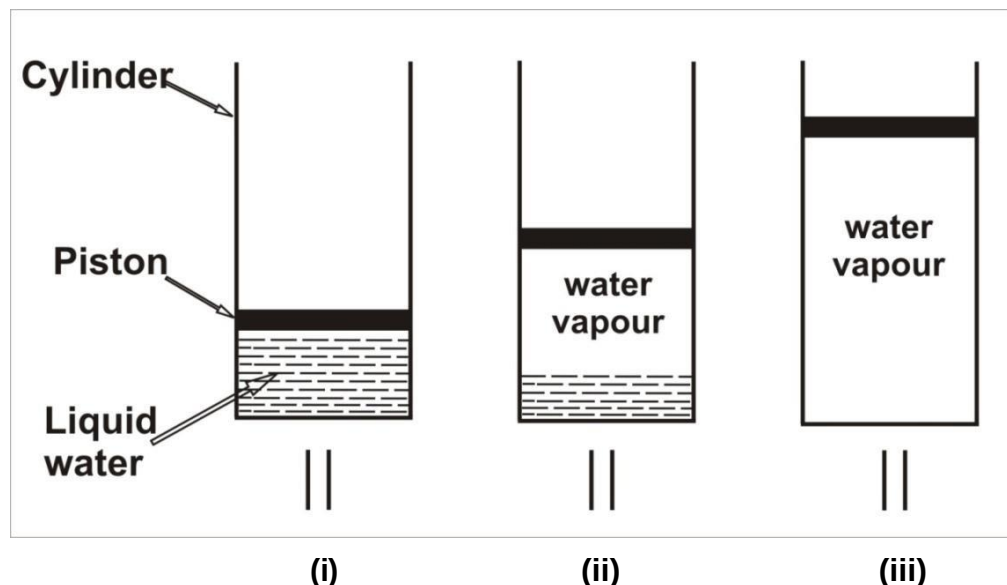
Note however that a uniform mixture of steam, hydrogen gas, and oxygen gas would be regarded as homogeneous in both composition and chemical aggregation whatever the relative proportions of the components.

_"*Invariable in chemical aggregation*" means that the state of chemical combination of the system does not change with *time* (condition (ii) referred to variation with *position*). Thus a mixture of hydrogen and oxygen, which changed into steam during the time that the system was under consideration, would not be a pure substance.

PHASE CHANGE OF A PURE SUBSTANCE

Let us consider 1kg of liquid water at a temperature of 20°C in a cylinder fitted with a piston, which exerts on the water a constant pressure of one atmosphere (1.0132 bar) as shown in Fig 2 (i).

As the water is heated slowly its temperature rises until the temperature of the liquid water becomes 100°C. During the process of heating, the *volume slightly increases* as indicated by the line 1-2 on the temperature – specific volume diagram (Fig.3). The piston starts moving upwards.



**Fig. 2 Phase change of water at constant pressure
From liquid to vapour phase.**

If the heating of the liquid, after it attains a temperature of 100°C, is continued it undergoes a change in phase. A portion of the liquid water changes into vapour as shown in Fig 2 (ii). This state is described by the line 2-3 in Fig. 3 The amount of heat

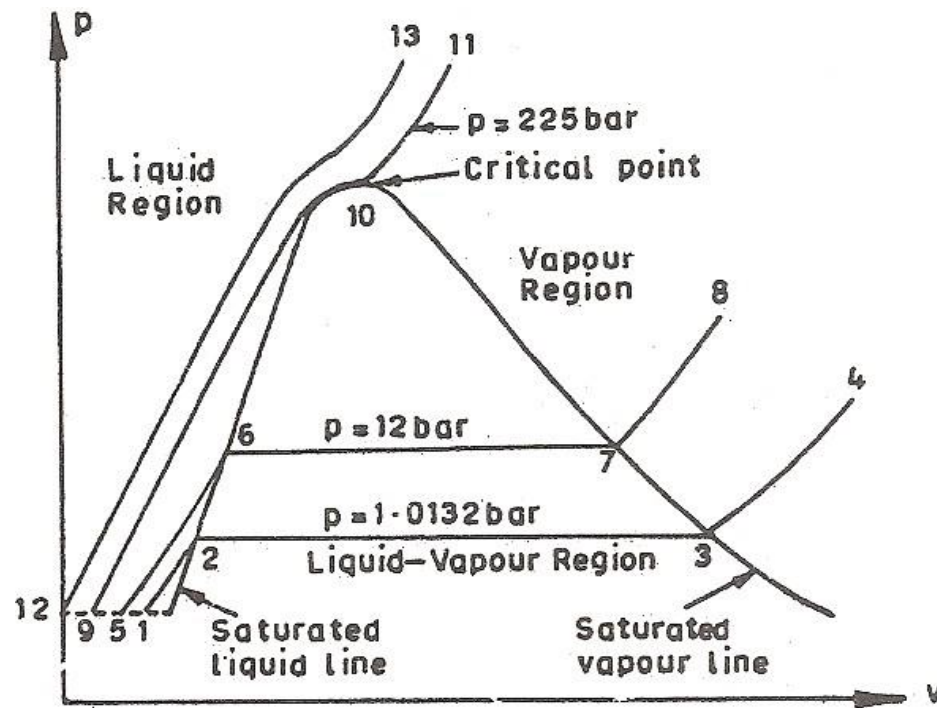


Fig.3

required to convert the liquid water completely into vapour under this condition is called the heat of vaporization. The temperature at which vaporization takes place at a given pressure is called the saturation temperature and the given pressure is called the saturation pressure. During the process represented by the line 2-3 (Fig.3) the volume increases rapidly and piston moves upwards Fig. 2 (iii).

For a pure substance, definite relationship exists between the saturation pressure and saturation temperature as shown in Fig. 4, the curve so obtained is called vapour pressure curve.

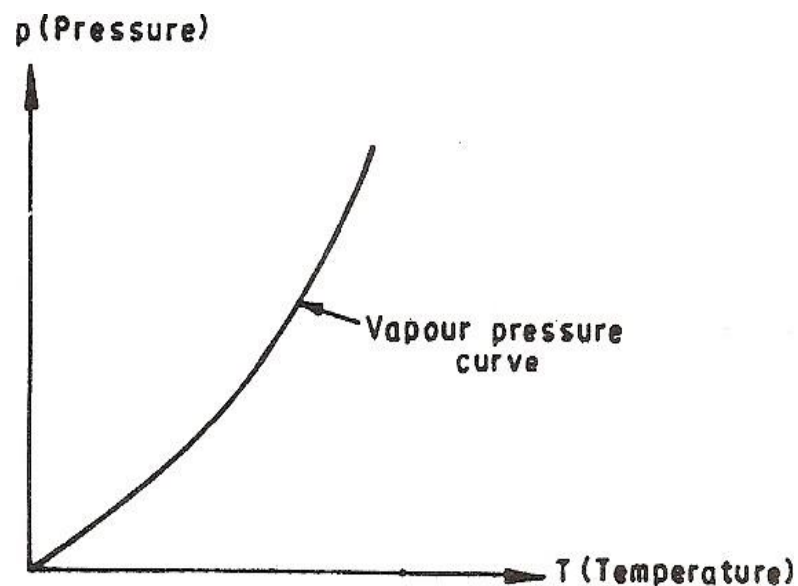


Fig. 4 Vapour pressure curve for water.

It may be noted that if the temperature of the liquid water on cooling becomes lower than the saturation temperature for the given pressure, the liquid water is called a sub cooled liquid. The point '1' (in Fig. 3.3) illustration, when the liquid water is cooled under atmospheric pressure to a temperature of 20°C, which is below the saturation temperature (100°C).

Further, at point '1' the temperature of liquid is 20°C and corresponding to this temperature, the saturation pressure is 0.0234 bar, which is lower than the pressure on the liquid water, which is 1 atmosphere. Thus the pressure on the liquid water is greater than the saturation pressure at a given temperature. In this condition, the liquid water is known as the compressed liquid.

The term compressed liquid or sub cooled liquid is used to distinguish it from saturated liquid. All points in the liquid region indicate the states of the compressed liquid.

When all the liquid has been evaporated completely and heat is further added, the temperature of the vapour increases. The curve 3-4 in Fig. 3 describes the process. When the temperature increases above the saturation temperature (in this case 100°C), the vapour is known as the superheated temperature. There is rapid increase in volume and the piston moves upwards [Fig. 2 (iii)].

The difference between the superheated temperature and the saturation temperature at the given pressure is called the degree of superheat.

If the above mentioned heating process is repeated at different pressures a number of curves similar to 1-2-3-4 are obtained. Thus, if the heating of the liquid water in the piston cylinder arrangement takes place under a constant pressure of 12 bar with an initial temperature of 20°C until the liquid water is converted into superheated steam, then curve 5-6-7-8 will represent the process.

In the above heating process, it may be noted that, as the pressure increases the length of constant temperature vaporizations gets reduced.

From the heating process at a constant pressure of 225 bar represented by the curve 9-10-11 in Fig. 3, it can be seen that there is no constant temperature vaporizations line. The specific volume of the saturated liquid and of the saturated vapour is the same, i.e. $= v_g$. Such states are called critical parameters.

The curve 12-13 (Fig. 3) represents a constant pressure heating process, when the pressure is greater than the critical pressure. At this state, the liquid water is directly converted into superheated steam. As there is no definite point at which the liquid water changes into superheated steam, it is generally called liquid water when the temperature is less than the critical temperature and superheated steam when the temperature is above the critical temperature.

P-T (Pressure – Temperature) DIAGRAM FOR A PURE SUBSTANCE

If the vapour pressure of a solid is measured at various temperatures until the triple point is reached and then that of the liquid is measured until the critical point is reached, the result when plotted on a P-T diagram appears as in Fig. 5.

If the substance at the triple point is compressed until there is no vapour left and the pressure on the resulting mixture of liquid and solid is increased, the temperature will have to be changed for equilibrium to exist between the solid and the liquid.

Measurements of these pressures and temperatures give rise to a third curve on the P-T diagram, starting at the triple point and continuing indefinitely.

The points representing the coexistence of (i) solid and vapour lie on the 'sublimation curve', (iii) liquid and solid lie on the 'fusion curve'. In the particular case of water, the sublimation curve is called the frost line, the vaporizations curve is called the steam line, and the fusion curve is called the ice line.

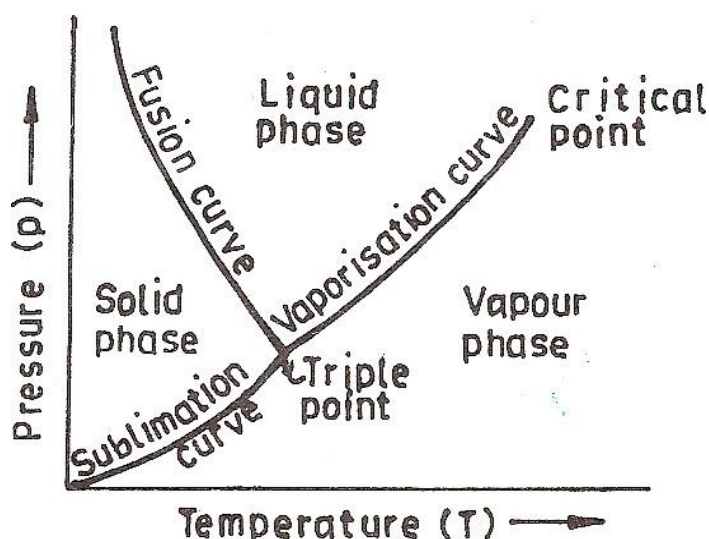


Fig .5 P-T diagram for a substance such as water

The slopes of sublimation and the vaporizations curves for all substances are positive. The slope of the fusion curve however may be positive or negative. The fusion curves of most substances have a positive slope. Water is one of the important exceptions.

Triple point

The triple point is merely the point of intersection of sublimation and vaporizations curves. It must be understood that only on P-T diagram is the triple point represented by a point. On P-V diagram it is a line, and on a U-V diagram it is a triangle.

The pressure and temperature at which all three phases of a pure substance coexist may be measured with the apparatus that is used to measure vapour pressure.

Triple-point data for some interesting substances are given in table 3.1.

Table 3.1 Triple-point Data

S. No.	Substance	Temp., K	Pressure, mm Hg
1.	Hydrogen (normal)	13.96	54.1
2.	Deuterium (normal)	18.63	128
3.	Neon	24.57	324
4.	Nitrogen	63.18	94
5.	Oxygen	54.36	1.14
6.	Ammonia	195.40	45.57
7.	Carbon dioxide	216.55	3,880
8.	Sulphur dioxide	197.68	1.256
9.	Water	273.16	4.58

P-V-T (Pressure-Volume-Temperature) SURFACE

A detailed study of the heating process reveals that the temperature of the solid rises and then during the change of phase from solid to liquid (or solid to vapour) the temperature remains constant. This phenomenon is common to all phase changes. Since the temperature is constant, pressure and temperature are not independent properties and cannot be used to specify state during a change of phase.

The combined picture of change of pressure, specific volume and temperature may be shown on a three dimensional state model. Fig 6, illustrate the equilibrium states for a pure substance which expands on fusion. Water is an example of a substance that exhibits this phenomenon.

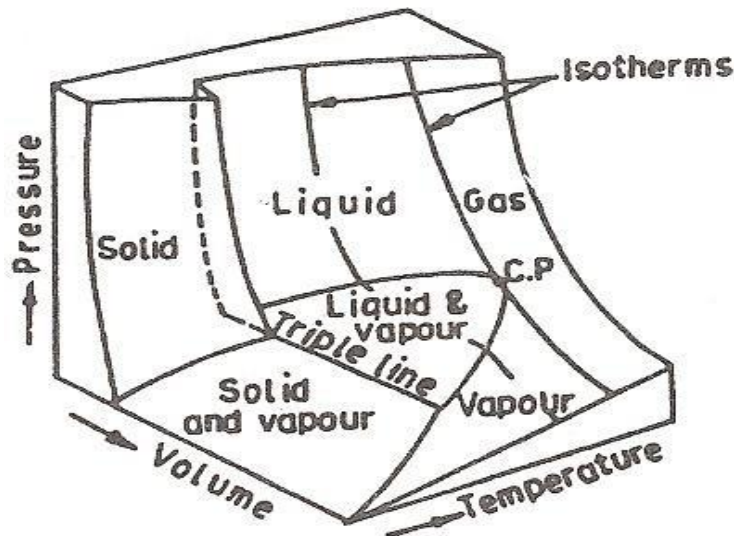


Fig.6 A Pressure-Volume-Temperature (P-V-T)

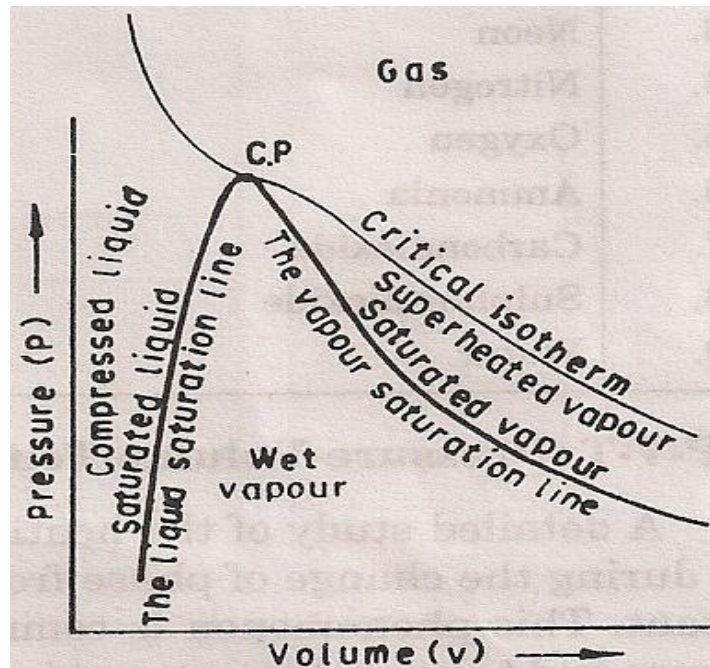
All the equilibrium states lie on the surface of the model. States represented by the space above or below the surface are not possible. It may be seen that the triple point appears as a line in this representation. The point C.P. is called the critical point and no liquid phase exists at temperatures above the isotherms through this point. The term evaporation is meaningless in this situation.

At the critical point the temperature and pressure are called the critical temperature and the critical pressure respectively and when the temperature of a substance is above the critical value, it is called a gas. It is not possible to cause a phase change in a gas unless the temperature is lowered to a value less than the critical temperature. Oxygen and nitrogen are examples of gases that have critical temperatures below normal atmospheric temperature.

PHASE CHANGE TERMINOLOGY AND DEFINITIONS

Suffices:	Solid	<i>i</i>
	Liquid	<i>f</i>
	Vapour	<i>g</i>

Phase change Process suffix	Name	Process
1. Solid-liquid <i>if</i>	Fusion	Freezing, melting
2. Solid-vapour <i>ig</i>	Sublimation	Frosting, defrosting
3. Liquid-vapour <i>fg</i>	Evaporation	Evaporating, Condensing



Phase change terminology.

Triple point: The only state at which the solid, liquid and vapour phases coexist in equilibrium.

Critical point (C.P): The limit of distinction between a liquid and vapour.

Critical pressure: The pressure at the critical point.

Critical temperature: The temperature at the critical point.

Gas – A vapour whose temperature is greater than the critical temperature.

Liquid-Vapour terms:

Saturation temperature: The phase change pressure.

Compressed liquid: Liquid whose temperature is lower than the saturation temperature. Sometimes called a sub cooled liquid.

Saturated liquid: Liquid at the saturation temperature corresponding to the saturation pressure. That is liquid about to commence evaporating, represented by the point f on a diagram.

Saturated vapour: A term including wet and dry vapour.

Dry (saturated) vapour: Vapour which has just completed evaporation. The pressure and temperature of the vapour are the saturation values. Dry vapour is represented by a point g on a diagram.

Wet vapour: The mixture of saturated liquid and dry vapour during the phase change.

Superheated vapour: Vapour whose temperature is greater than the saturation temperature corresponding to the pressure of the vapour.

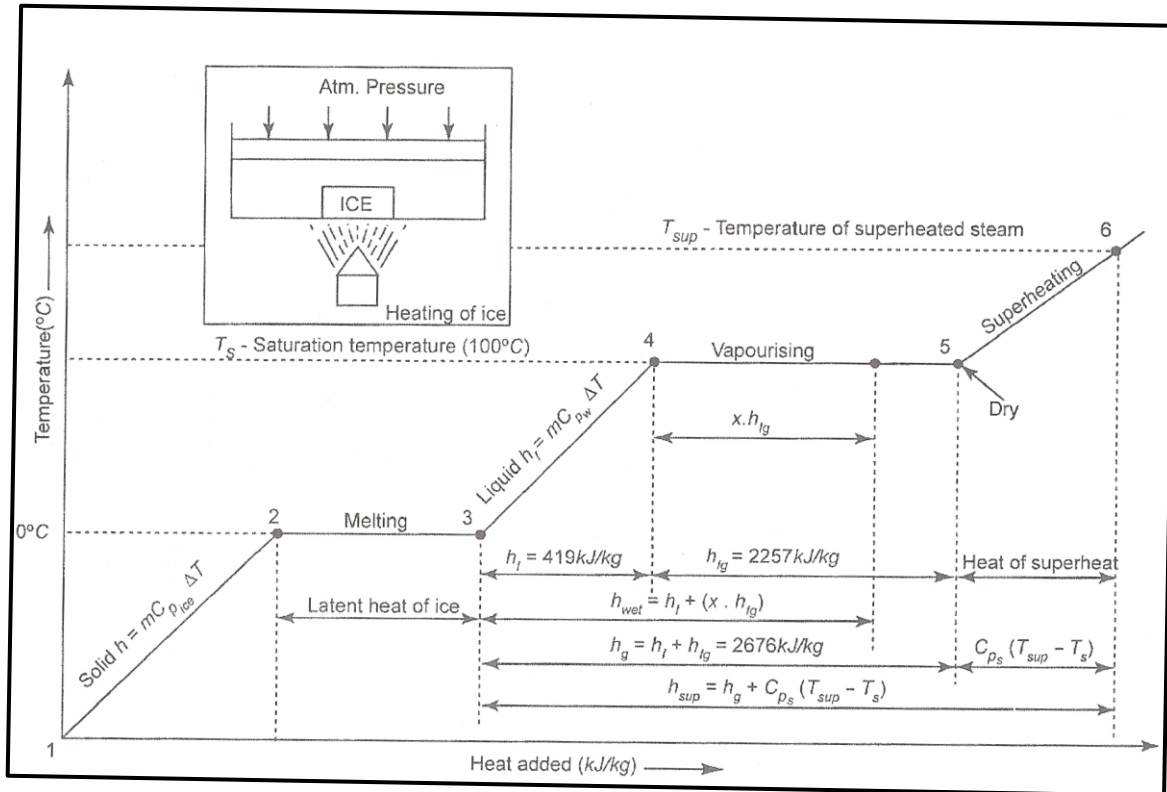
Degree of superheat: The term used for the numerical amount by which the temperature of a superheated vapour exceeds the saturation temperature.

FORMATION OF STEAM AND ITS PROPERTIES

INTRODUCTION

For any system we need a working substance to convert a heat energy which is produced by a fuel into mechanical work. In the previous article we considered an ideal gas as a working substance. But in practice, there is no ideal gases exist. In this article, we consider steam as a working substance because of its following unique properties

- Steam can be produced from water which is cheap and readily available in large quantity.
- It can carry large amount of heat.
- It can be used for some other purposed after its duty as a working substance is completed.



STEAM FORMATION

Consider one kg of water in a closed vessel under a pressure of $p \text{ N/m}^2$ and at a temperature of -20°C . If we heat the water gradually when pressure remains constant, the following changes will occur:

- The temperature of the ice will increase till it reaches the freezing temperature of water i.e. 0°C . This is shown by the line 1-2 in fig.
- When we add more heat after the point 2, the ice starts melting at the same time. There is no rise in temperature till the whole of the ice has been melted and converted into water. This process is represented by a line 2-3 in fig. The heat added during this period is called **Latent heat of fusion of ice** or **Latent heat of ice**.
- On further heating, the water reaches its boiling point or saturation point 4. This saturation temperature varies with pressure. At the atmospheric pressure, boiling point of water is 100°C . The amount of heat added during heating of water from 0°C to saturation temperature of 100°C is known as **sensible heat or Latent heat of water**. It is denoted by h_f .

$$\text{Mathematically, } h_f = m C_p (T_2 - T_1)$$

- (d) On further heating beyond 4, the water will gradually be converted into steam, when the temperature remains constant. At this stage, the steam will have some water particles in suspension and is called **wet steam**. The same process continues till all the water particles converted into wet steam. The line 4-5 in fig. represents this process.
- (e) If the water is heated further the water particles in suspension will be converted into steam. The same without any water suspension is called as **dry steam or dry saturated steam**.

The amount of heat added during heating of water from boiling point to dry saturated stage is called as **Latent heat of vaporization or Enthalpy of vaporization or Latent heat of steam**. It is denoted by ' h_{fg} '.

- (f) When the dry steam is further heated, the temperature rises again. This process is called '**Superheating**' and the steam obtained is known as **superheated Steam**.

The heat supplied to the dry steam at saturation temperature T_s (i.e. 100°C), to convert it into superheated steam at the temperature T_{sup} is called **heat of super heat of Superheat enthalpy**. It is denoted by ' h_{sup} '.

$$\text{Mathematically, } h_{sup} = h_g + C_p (T_{sup} - T_s) \text{ kJ/kg}$$

Dryness Fraction

It is defined as the ratio of the mass of the dry steam to the mass of wet steam actually Present. It is denoted by ' x '.

$$\text{Dryness fraction, } X = \frac{m_g}{m_f + m_g}$$

where,

m_g – mass of dry steam in kg

m_f – mass of water vapour in suspension.

This dryness fraction when expressed in percentage (i.e $100x$) is called the **quality of system**.

PROPERTIES OF STEAM

Various properties of water at various conditions of steam.

1. Enthalpy of steam(h):

It is the amount of heat added to the water from freezing point to till the water becomes wet of dry or superheated steam.

For wet steam, $h_{\text{wet}} = h_f + xh_{fg}$ kJ/kg

For dry steam, $h_{\text{dry}} = h_g = h_f + h_{fg}$ kJ/kg

For superheated steam, $h_{\text{sup}} = h_g + C_p(T_{\text{sup}} - T_{\text{sat}})$ kJ/kg

Where,

$(T_{\text{sup}} - T_{\text{sat}})$ is called as 'degree of superheat'.

2. Specific volume of steam (v):

It is defined as the volume occupied by the unit mass of the steam at the given pressure and temperature.

For wet steam, $v_{\text{wet}} = xv_g$ m³/kg

For dry steam, $v_{\text{dry}} = v_g$ m³/kg

For superheated steam, $v_{\text{sup}} = v_g T_{\text{sup}}/T_{\text{sat}}$ m³/kg

3. Density of steam (ρ):

It is defined as the ratio of mass to the unit volume of the steam at given pressure and temperature. Its value for wet, dry and superheated steam is the reciprocal of the specific volume of the steam.

4. Workdone during expansion(W):

During the evaporation process, there is a considerable increase in the volume when pressure remains constant. Thus some amount of external work is done during evaporation. In other words, it is the energy required for absorption of latent heat for increasing volume of the steam.

For wet steam, $W_{\text{wet}} = 100 Pxv_g$ kJ

For dry steam, $W_{\text{dry}} = 100Pv_g$ kJ

For superheated steam, $W_{\text{sup}} = 100 P v_{\text{sup}}$ kJ

Where p- pressure at which evaporation takes place in bar.

5. Internal energy of steam (U):

Internal energy of steam is defined as the actual heat energy stored in the steam above the freezing point of water at the given conditions. It is the difference between enthalpy of steam and the external work done.

$$h = W + \Delta u$$

$$\Delta u = h - W$$

For wet steam, $u_{\text{wet}} = (h_f + x h_{fg}) - (100 p x v_g)$ kJ/kg

For dry steam, $u_{\text{dry}} = (h_f + h_{fg}) - (100 p v_g)$ kJ/kg

For superheated steam, $u_{\text{sup}} = h_{\text{sup}} - (100 p v_{\text{sup}})$ kJ/kg

6. Entropy of steam(s):

It is the property of the steam which increases with increase in temperature and decrease with decrease in temperature.

For wet steam, $s_{\text{wet}} = s_f + x s_{fg}$ kJ/kg K

For dry steam, $s_{\text{dry}} = s_f + s_{fg}$ kJ/kg K

For superheated steam, $s_{\text{sup}} = s_g + C_{ps} \log_e(T_{\text{sup}}/T_s)$ kJ/kg K

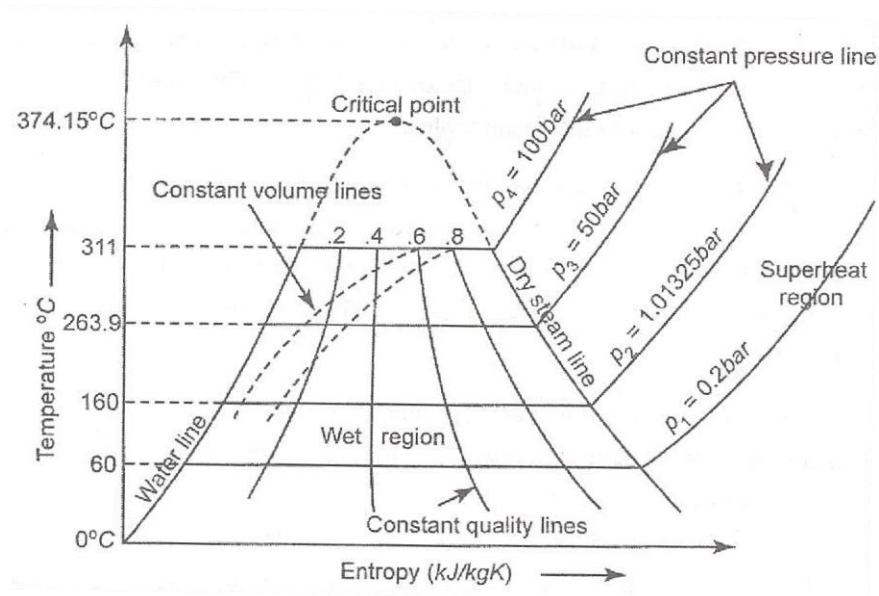
T – S DIAGRAM FOR WATER AND STEAM

T-s is the plot of the saturation temperature of water and steam corresponding to the various absolute against the entropies at those saturation temperatures. It is shown in fig.. It is very much useful to solve the numerical problems on steam as it considerably reduces time and labour in making calculations.

In the T-s diagram, the region left of the water line, the water exists as liquid. In the right of the dry steam line, the water exists as a wet steam. Therefore the dryness fraction lines are represented in these regions as shown in fig. 3.2. The value of various quantities can be directly read from the diagram.

It can be noted from the figure that the water line and steam line are converging with the increase in temperature. At a particular point, the water is directly converted into dry steam without formation of wet steam. This point is called '**critical Point**'.

The **critical temperature** is the temperature above which a substance can not exist as a liquid. The corresponding pressure is called **critical pressure**. The critical pressure and temperature of water are 221.2 bar and 374.15°C respectively.



An isothermal process is represented by a horizontal line and a reversible adiabatic or isentropic is represented by a vertical line as shown in fig. In the diagram, constant volume lines are plotted in the wet region and constant pressure lines are plotted in the superheated region.

The T - s diagram represents the whole cycle of heat addition and rejection whereas the p - V diagram represents only the work done.

USE OF STEAM TABLES

In practice, it is very tedious to calculate the relations between various quantities like pressure, temperature, specific volume, specific volume, enthalpy, and entropy of steam at various stages. These properties have been determined experimentally and tabulated in a specific manner is known as *steam tables*. The values of the various quantities given in these tables are for 1 kg of steam which is dry steam which is dry saturated. The calculation for wet steam can be easily made by making suitable use of such table. The following three main divisions present in the steam tables.

Table I shows the various quantities like specific volume, specific enthalpy, and specific entropy for various pressures. If the required pressure is not directly included in a steam tables it is obvious that it would be lying between two consecutive pressures. Such a data can be calculated by simple interpolation.

Table II shows the various quantities for various saturation temperatures since there is only one saturation temperature for each saturation pressure.

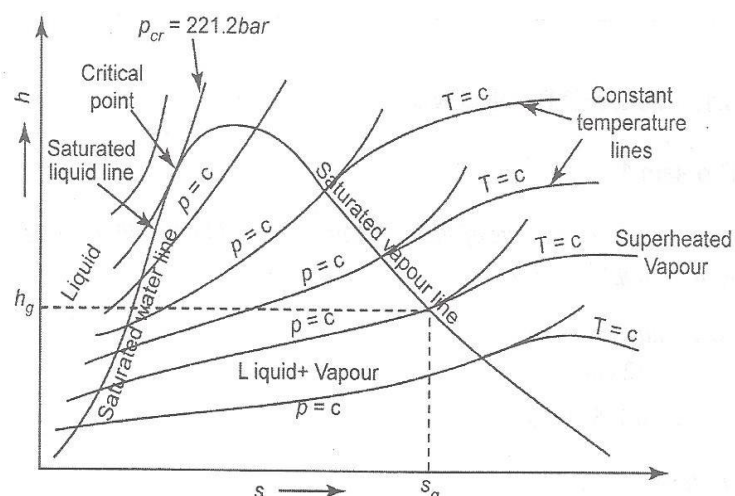
Table III shows the superheated steam table. Here the various quantities like specific volume, specific enthalpy and specific entropy for various pressure and temperature are shown in tabular form.

It can be noted from the properties of a saturated steam that the increase of pressure results in gradual increase of sensible heat and decrease of latent heat of vaporization. But the enthalpy of dry steam increases up to a certain pressure (approximately 33.5 bar) and then it decreases.

USE OF MOLLIER CHART

It is also called as *Enthalpy-Entropy ($h-s$) diagram*. In Mollier chart, the vertical ordinate represents the total heat while the base represents the entropy as shown in fig

In Mollier chart, the dry steam line divides this chart into two regions. The region which is below the dry steam line represents the wet condition of steam. Here, the dryness fraction lines are shown parallel to dry steam line. The region above the dry steam line represents the superheated condition of steam. In this region, constant temperature lines are shown. It should be noted that the lines at constant pressure are straight in the wet steam region but curved in the superheated region.



PROBLEMS:

1. Find the saturation temperature, the changes in specific volume and entropy during evaporation, and the latent heat of vaporization of steam at 1 MPa.

Solution At 1 MPa, from steam table

$$t_{\text{sat}} = 179.91^\circ\text{C}$$

$$v_f = 0.001127 \text{ m}^3/\text{kg}$$

Ans.

$$v_g = 0.19444 \text{ m}^3/\text{kg}$$

$$\therefore v_{fg} = v_g - v_f = 0.1933 \text{ m}^3/\text{kg}$$

$$s_f = 2.1387 \text{ kJ/kg K}$$

$$s_g = 6.5865 \text{ kJ/kg K}$$

$$\therefore s_{fg} = s_g - s_f = 4.4478 \text{ kJ/kg K}$$

Ans.

$$h_{fg} = h_g - h_f = 2015.3 \text{ kJ/kg}$$

Ans.

2. Find the enthalpy, entropy, and volume of steam at 1.4 MPa, 380°C.

Solution At $p = 1.4 \text{ MPa}$, in Table A.1(b), $t_{\text{sat}} = 195.07^\circ\text{C}$. Therefore, the state of steam must be in the superheated region. In Table A.2, for properties of superheated steam,

at 1.4 MPa, 350°C

$$v = 0.2003 \text{ m}^3/\text{kg}$$

$$h = 3149.5 \text{ kJ/kg}$$

$$s = 7.1360 \text{ kJ/kg K}$$

and at 1.4 MPa, 400°C

$$v = 0.2178 \text{ m}^3/\text{kg}$$

$$h = 3257.5 \text{ kJ/kg}$$

$$s = 7.3026 \text{ kJ/kg K}$$

$\therefore$ By interpolation

at 1.4 MPa, 380°C

$$v = 0.2108 \text{ m}^3/\text{kg}$$

$$h = 3214.3 \text{ kJ/kg}$$

$$s = 7.2360 \text{ kJ/kg K}$$

Ans.

3. A vessel of volume 0.04 m^3 contains a mixture of saturated water and saturated steam at a temperature of 250°C . The mass of the liquid present is 9 kg . Find the pressure, the mass, the specific volume, the enthalpy, the entropy, and the internal energy.

Solution From Table A.1(a), at 250°C $p_{\text{sat}} = 3.973 \text{ MPa}$
 $v_f = 0.0012512 \text{ m}^3/\text{kg}$, $v_g = 0.05013 \text{ m}^3/\text{kg}$
 $h_f = 1085.36 \text{ kJ/kg}$, $h_{fg} = 1716.2 \text{ kJ/kg}$
 $s_f = 2.7927 \text{ kJ/kg K}$, $s_{fg} = 3.2802 \text{ kJ/kg K}$

Volume of liquid, $V_f = m_f v_f$
 $= 9 \times 0.0012512$
 $= 0.01126 \text{ m}^3$

Volume of vapour, $V_g = 0.04 - 0.01126$
 $= 0.02874 \text{ m}^3$

$\therefore$ Mass of vapour

$$m_g = \frac{V_g}{v_g} = \frac{0.02874}{0.05013} = 0.575 \text{ kg}$$

$\therefore$ Total mass of mixture,

$$m = m_f + m_g = 9 + 0.575 = 9.575 \text{ kg} \quad \text{Ans.}$$

Quality of mixture,

$$x = \frac{m_g}{m_f + m_g} = \frac{0.575}{9.575} = 0.06$$

$\therefore$ $v = v_f + xv_{fg}$
 $= 0.0012512 + 0.06(0.05013 - 0.0012512)$
 $= 0.00418 \text{ m}^3/\text{kg} \quad \text{Ans.}$

$$h = h_f + xh_{fg}$$

$$= 1085.36 + 0.06 \times 1716.2$$

$$= 1188.32 \text{ kJ/kg} \quad \text{Ans.}$$

$$s = s_f + xs_{fg}$$

$$= 2.7927 + 0.06 \times 3.2802$$

$$= 2.9895 \text{ kJ/kg K} \quad \text{Ans.}$$

$$u = h - pv$$

$$= 1188.32 - 3.973 \times 10^3 \times 0.00418$$

$$= 1171.72 \text{ kJ/kg} \quad \text{Ans.}$$

Also, at 250°C , $u_f = 1080.39$ and $u_{fg} = 1522.0 \text{ kJ/kg}$

$\therefore$ $u = u_f + xu_{fg}$
 $= 1080.39 + 0.06 \times 1522$
 $= 1071.71 \text{ kJ/kg} \quad \text{Ans.}$

4. Steam initially at 0.3 MPa, 250°C is cooled at constant volume.

(a) At what temperature will the steam become saturated vapour? (b) What is the quality at 80°C? What is the heat transferred per kg of steam in cooling from 250°C to 80°C?

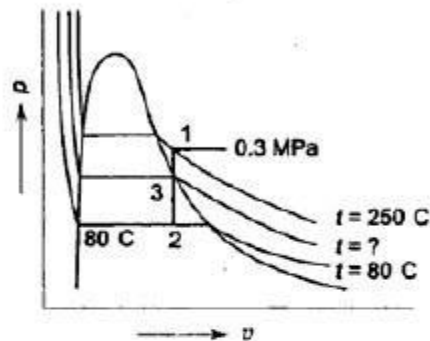
Solution At 0.3 MPa, $t_{\text{sat}} = 133.55^\circ\text{C}$

Since $t > t_{\text{sat}}$, the state would be in the superheated region (Fig)

Properties of superheated steam, at 0.3 MPa, 250°C

$$v = 0.7964 \text{ m}^3/\text{kg}$$

$$h = 2967.6 \text{ kJ/kg}$$



$$\therefore v_1 = v_3 = v_2 = 0.7964 \text{ m}^3/\text{kg}$$

In Table A.1

$$\text{when } v_g = 0.8919, t_{\text{sat}} = 120^\circ\text{C}$$

$$\text{when } v_g = 0.7706, t_{\text{sat}} = 125^\circ\text{C}$$

Therefore, when $v_g = 0.7964$, t_{sat} , by linear interpolation, would be 123.9° .

Steam would become saturated vapour at $t = 123.9^\circ\text{C}$

Ans. (a)

At 80°C , $v_f = 0.001029 \text{ m}^3/\text{kg}$, $v_g = 3.407 \text{ m}^3/\text{kg}$,

$$h_f = 334.91 \text{ kJ/kg}, h_{fg} = 2308.8 \text{ kJ/kg}, p_{\text{sat}} = 47.39 \text{ kPa}$$

$$\begin{aligned} v_1 = v_2 = 0.7964 \text{ m}^3/\text{kg} &= v_{f80^\circ\text{C}} + x_2 v_{fg80^\circ\text{C}} \\ &= 0.001029 + x_2 (3.407 - 0.001029) \end{aligned}$$

$$\therefore x_2 = \frac{0.79539}{3.40597} = 0.234 \quad \text{Ans. (b)}$$

$$h_2 = 334.91 + 0.234 \times 2308.8 = 875.9 \text{ kJ/kg}$$

$$h_2 = 2967.6 \text{ kJ/kg}$$

From the first law of thermodynamics

$$dQ = du + p dv$$

$$\therefore (dQ)_v = du$$

$$\begin{aligned} \text{or } Q_{1-2} &= u_2 - u_1 = (h_2 - p_2 v_2) - (h_1 - p_1 v_1) \\ &= (h_2 - h_1) + v(p_1 - p_2) \\ &= 875.9 - 2967.6 + 0.7964 (300 - 47.39) \\ &= 2091.7 + 201.5 \\ &= -1890.2 \text{ kJ/kg} \end{aligned}$$

Ans. (c)

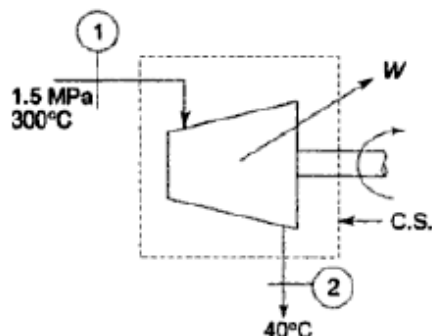
5. Steam initially at 1.5 MPa, 300°C expands reversibly and adiabatically in a steam turbine to 40°C. Determine the ideal work output of the turbine per kg of steam.

Solution The steady flow energy equation for the control volume, as shown in Fig. gives (other energy terms being neglected)

$$h_1 = h_2 + W$$

∴

$$W = h_1 - h_2$$



Work is done by steam at the expense of a fall in its enthalpy value. The process is reversible and adiabatic, so it is isentropic. The process is shown on the T - s and h - s diagrams

From Table A.1(a), at 40°C

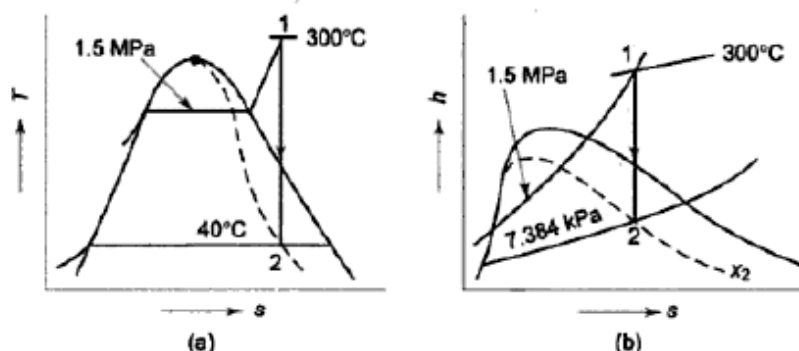
$$p_{\text{sat}} = 7.384 \text{ kPa}, s_f = 0.5725, \text{ and } s_{fg} = 7.6845 \text{ kJ/kg K}$$

$$h_f = 167.57, \text{ and } h_{fg} = 2406.7 \text{ kJ/kg}$$

At $p = 1.5 \text{ MPa}$, $t = 300^\circ\text{C}$, from the tabulated properties of superheated steam (Table A.2)

$$s_1 = 6.9189 \text{ kJ/kg K}$$

$$h_1 = 3037.6 \text{ kJ/kg}$$



Since

$$s_1 = s_2$$

$$6.9189 = s_f + x_2 s_{fg, 40^\circ\text{C}}$$

$$= 0.5725 + x_2 \times 7.6845$$

$$x_2 = \frac{6.3464}{7.6845} = 0.826 \text{ or } 82.6\%$$

∴

$$h_2 = h_{f, 40^\circ\text{C}} + x_2 h_{fg, 40^\circ\text{C}}$$

$$= 167.57 + 0.826 \times 2406.7$$

$$= 2152.57 \text{ kJ/kg}$$

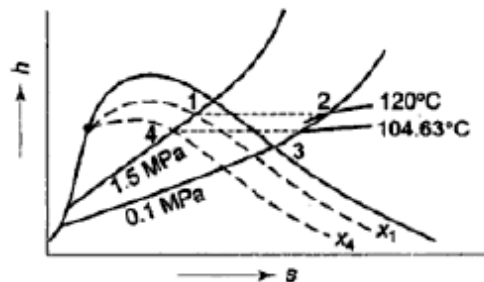
∴

$$W = h_1 - h_2 = 3037.6 - 2152.57$$

$$= 885.03 \text{ kJ/kg}$$

Ans.

Solution At state 2 when $p = 0.1$ MPa, $t = 120^\circ\text{C}$ by inter-polation



$$h_1 = h_2$$

$$h_{f1.5\text{MPa}} + x_1 h_{fg1.5\text{MPa}} = h_2$$

$$844.89 + x_1 \times 1947.3 = 2716.2$$

$$x_1 = \frac{1871.3}{1947.3} = 0.963$$

Ans.

When $p = 0.1$ MPa and $t = 99.63 + 5 = 104.63^\circ\text{C}$

$$h_3 = 2685.5 \text{ kJ/kg}$$

$$h_3 = h_4$$

$$2685.5 = 844.89 + x_4 \times 1947.3$$

$$x_4 = \frac{1840.6}{1947.3} = 0.948$$

The maximum moisture that can be determined with this set-up is only 5.2%.

Ans.